Algebra Monsoon 2026 Lecture 9 Groups acting on sets

Groups acting on sets

Let (M, μ, e) be a monoid.

Recall that BM is the category defined by

- $(BM)_0 = \{M\}$,
- $(BM)_1 = M$,
- s_0, s_1 are both equal to the unique map $M \longrightarrow \{M\}$,
- $Id_M = e$
- $f \circ g = \mu(f, g)$ for $f, g \in M$.

Def. An M-set is a functor

$$F : BM \longrightarrow \mathcal{E}.$$

(where $\mathcal{E}$ is the category of sets).

Let's unpack this definition.

Functors are determined by what they do to objects and morphisms.

So to specify F, we have give

- a set $S = F(G)$ in $\mathcal{E}_0$
- a morphism of monoids

$$\varphi : M \longrightarrow \mathcal{E}(S,S)$$

Under the adjunction

$$\mathcal{E}\left(M, \mathcal{E}(S,S)\right) \xrightarrow{\sim} \mathcal{E}\left(M \times S, S\right),$$

the morphism φ corresponds to a mapping $\alpha : M \times S \longrightarrow S$ satisfying

- $\alpha(e, s) = s$ for all $s \in S$
- $\alpha(n, \alpha(m, s)) = \alpha(nm, s)$

for all $s \in S$ and $m, n \in M$.

Conversely, given a set S and a mapping $M \times S \xrightarrow{\alpha} S$ satisfying these two conditions, there is a unique G-set $F : BM \longrightarrow \mathcal{E}$ such that

- $F(M) = S$

- the morphism of monoids $M \to \mathcal{E}(S,S)$ corresponds under adjunction to α.

So an M-set is a set together with a mapping $M \times S \longrightarrow S$ satisfying

- $e.s = s$ for all $s \in S$
- $n.(m.s) = nm.s$

for all $s \in S$ and $m, n \in M$.

The notion of morphisms of functors gives the following definition

Def. A morphism of M-sets $S \to T$ is a mapping $f : S \to T$ such that

$$f(m.s) = m.f(s)$$

for every $m \in M$ and every $s \in S$.

Exercise Show that this is a translation of the notion of morphisms of functors.

The category of M-sets

Let M be a monoid.

We have the category $\mathcal{E}^{BM}$ of M-sets and their morphisms.

Let $f : N \longrightarrow M$ be a morphism of monoids. We have a natural functor

$$\mathcal{E}^{BM} \longrightarrow \mathcal{E}^{BN}$$

$$F \longmapsto F \circ I$$

where $I : BN \longrightarrow BM$ is the functor corresponding to $f : N \to M$.

For us, the monoid M will be a group G.

Example Let G be a group,

We can view G as an G-set under the mapping

$$G \times G \longrightarrow G$$
$$(h, g) \longmapsto hg$$

(because $eg = g$ and $h_2(h_1 g) = (h_2 h_1)g$).

This is called the action of the group G on the set G by left translations.

The corresponding morphism of monoids

$$G \longrightarrow End_{\mathcal{E}}(G)$$

is $$h \longmapsto \gamma_h$$

(recall $\gamma_e = Id_G$ and $\gamma_{h_2} \circ \gamma_{h_1} = \gamma_{h_2 h_1}$).

Exercise: Let G be a group, E a G-set, and F any set. For $g \in G$ and $f \in \mathcal{E}(E,F)$, define $g.f \in \mathcal{E}(E,F)$ by $(g.f)(x) = f(g^{-1}.x)$. Show that this makes $\mathcal{E}(E,F)$ into a G-set.

Orbits

Let G be a group and S a G-set.

For $s \in S$, the subset

$$G.s = \{ g.s \mid g \in G \}$$

of S is called the G-orbit of s; the set of G-orbits of elements of S is denoted $G \backslash S$.

- For $s \in S$, the mapping $G \to S$, $g \mapsto g.s$ is called the orbital mapping at x. The image of this mapping is $G.s$.

Lemma If $t \in G.s$, then $s \in G.t$.

Proof: Indeed, if $t = g.s$ for some $g \in G$, then

$$\begin{aligned} \bar{g}^1.t &= \bar{g}^1.(g.s) \\ &= (\bar{g}^1 g).s \\ &= e.s \\ &= s, \end{aligned}$$

so $s \in G.t$. □

Lemma If $t \in G.s$, then

$$G.t = G.s.$$

Proof Suppose that $t = g.s$ for some $g \in G$. For every $h \in G$, we have

$$h.t = h.(g.s) = (hg).s$$

showing that $G.t \subset G.s$.
We have seen that if $t \in G.s$, then $s \in G.t$. Interchanging the roles of s and t, we get $G.s \subset G.t$. Hence $G.t = G.s$. □

Let G be a group and S a G-set.

Proposition The set of G-orbits $G\backslash S$ is a partition of S.

Proof i) No G-orbit $G.s$ is empty because $s \in G.s$; indeed $s = e.s$.

ii) $\forall t \in G.s \cap G.t$ for some

ii) If $s, t, u \in S$, then

$$G.s = G.u \text{ and } G.t = G.u$$

by the above lemmas, so

$$G.s = G.t.$$

iii) Clearly, S is the union of all G-orbits in S because each $s \in S$ belongs to $G.s$.

It follows that the G-orbits form a partition of S. □

Let G be a group and S a G-set.

Proposition The equivalence relation $\sim$ on S corresponding to the partition $G\backslash S$ of S is

$$s \sim t \iff G.s = G.t.$$

The partition of S and $G\backslash S$,

Proof By the definition of $\sim$ we have $s \sim t$ if and only if there exists $u \in S$ such that

$$s, t \in G.u.$$

This implies $G.u = G.s$ and $G.u = G.t$ (by the lemmas), so $G.s = G.t$. Conversely, if $G.s = G.t$ for some $s, t \in S$, then obviously $s \sim t$ because $s \in G.s$ and $t \in G.t$. $\square$